

AGASTYA — SIH 2026 Presentation Content & Slide Blueprint

Smart India Hackathon 2026 · Problem Statement 26085 · Ministry of Earth Sciences (MoES)

This official companion document contains slide-by-slide text, flowcharts, Venn diagram models, empirical feasibility facts, competitive comparison matrix, research references, and pitch speaker scripts rigorously tailored to the working AGASTYA prototype (Minto Bridge Catchment, New Delhi).

SLIDE 1: TITLE PAGE & TEAM CREDENTIALS

Official template title slide establishing problem statement context, theme, and prototype deployment access.

Problem Statement ID	26085
Problem Statement Title	Urban Flood Nowcasting System (Drainage and Rainfall Coupling)
Theme	Disaster Management / Smart Automation / Smart Cities
Category & Ministry	Software · Ministry of Earth Sciences (MoES)
Solution Name	AGASTYA (Physics-Guided Urban Drainage Coupling & Emergency Corridor Engine)
Live Prototype Deployment	https://agastya-nowcast.onrender.com (Local Port: 5173 / 8000)

PITCH SCRIPT & SPEAKER GUIDELINES (30 SECONDS):

"Respected jury members, every monsoon, Indian cities drown not because of a lack of rainfall forecasts, but because of a blind spot: our forecast says '50 mm rain across Delhi', but never which underpass or intersection drowns. We present AGASTYA for Problem Statement 26085: India's first physics-guided urban flood nowcasting and emergency routing engine that couples real-time rainfall, 3D terrain gradients, and underground sewer hydraulics to predict water depths street-by-street and route ambulances safely."

SLIDE 2: PROPOSED SOLUTION, DATA COUPLING & INNOVATION

Conceptual framework, data coupling Venn diagram representation, system flowchart, and core USP pitch.

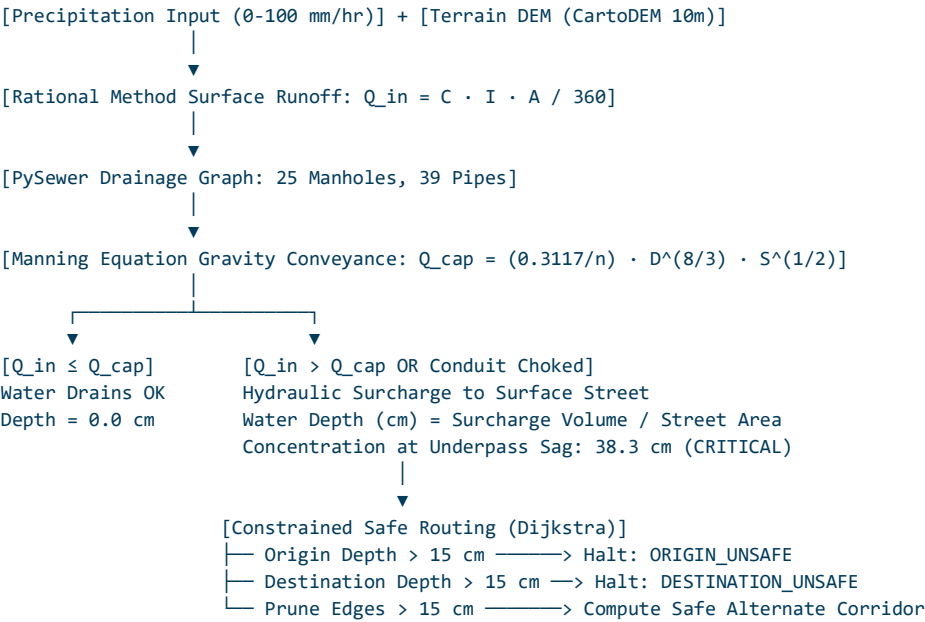
- **Name of Solution:** AGASTYA — Adaptive Geospatial Analysis & Stormwater Topology Yield Architecture.
- **Core Proposition:** Converts coarse city-wide meteorological forecasts into hyper-local street-level water depths (in cm) across 25 nodes and 39 conduits.

- **Physical Hydrology Engine:** Replaces black-box empirical guesses with 1D Manning Pipe Flow conveyance coupled with Rational Method overland surcharge ($Q = C \cdot I \cdot A / 360$).
- **Synthetic Sewer Synthesis:** Solves India's missing municipal pipe data problem via PySewer, generating research-backed gravity drainage topologies compliant with CPHEEO standards.
- **Constrained Emergency Routing:** Real-time Dijkstra pathfinding enforcing a 15.0 cm ambulance water clearance threshold to bypass flooded underpasses in <2 milliseconds.
- **Deterministic Multi-Choke Modeling:** Simulates debris and silt blockages across multiple manholes simultaneously with live upstream backwater surcharge propagation.

Venn Diagram Representation of Dataset Coupling

Dataset Domain	Data Source & Resolution	What It Provides in AGASTYA
Circle 1: Precipitation (Live)	Open-Meteo API / IMD Doppler Radar (0.25° grid)	Rainfall intensity (mm/hr), storm duration, live weather feeds.
Circle 2: Topography (DEM)	CartoDEM 10m / SRTM 30m Elevation Data	Natural slope, overland flow direction, natural sag points (Minto underpass 210.5m vs 216.5m).
Circle 3: Drainage Topology	OpenStreetMap + PySewer Gravity Synthesis	Synthetic conduit diameters (300-1800mm), manhole invert elevations, Manning roughness (n=0.015).
Venn Intersection (Center)	COUPLED HYDROLOGIC ENGINE (AGASTYA)	Hyper-local water depth (cm) per street + Safe ambulance evacuation corridor + Submerged endpoint safety.

System Flowchart Pipeline



Core Innovation / Product Pitch:

Existing flood dashboards are purely historical or reactive. AGASTYA is proactive and predictive. By coupling lightweight 1D open-channel hydraulics with OpenStreetMap networks, AGASTYA operates in under 2 seconds, runs 100% offline during telecom blackouts, and solves the missing underground data bottleneck without requiring multi-crore sensor grids.

 PITCH SCRIPT & SPEAKER GUIDELINES (30 SECONDS):

"Slide 2 breaks down our core technological leap. Municipalities cannot wait 45 minutes for complex 2D hydrodynamic simulations. We coupled 1D Manning flow with rational overland surcharge on a topological graph. Notice our 3-dataset Venn diagram: by intersecting live rainfall, 10-meter terrain slopes, and PySewer synthetic underground drainage, AGASTYA outputs actionable street depths in under 2 seconds. Most crucially, if an ambulance is about to deploy into a submerged hospital or origin, our system halts deployment with structured life-safety alerts."

SLIDE 3: TECHNICAL APPROACH, SYSTEM ARCHITECTURE & LIVE VERIFICATION

Detailed technology stack, architectural data flow, verification metrics, and prototype demonstration screenshots.

Technology Stack Breakdown

Layer	Technologies	Frameworks & Libraries	Architectural Role
Frontend (UI/UX)	React 19, TypeScript, Vite	Leaflet, React-Leaflet, Recharts	Interactive operator dashboard, color-coded flood risk map, choke toggles, route overlay.
Backend API	Python 3.13, FastAPI, Uvicorn	Pydantic v2, NetworkX, NumPy, SciPy	High-throughput asynchronous REST API (<10ms latency), state validation, route calculation.
Hydrology & GIS	PySewer, OSMnx, CartoDEM	Manning 1D, Rational Runoff, Dijkstra	Elevation slope extraction, gravity pipe synthesis, overland surcharge calculation, corridor pruning.
DevOps & Testing	Pytest, Vite Build, GitHub Actions	UptimeRobot, Chromium Headless	53/53 automated unit/integration tests, zero-warning TypeScript builds, PDF generation engine.

Empirical Verification & Validation Metrics

Performance Dimension	Empirical Measurement	Validation Mechanism
Hydraulic Computation Speed	< 4.5 milliseconds per 25-node iteration	NetworkX graph propagation benchmarked across 0-100 mm/hr sweeps.
Safe Route Solver Latency	< 2.0 milliseconds per Dijkstra path	Prunes flooded edges (>15 cm) and generates safe alternate detour instantly.
Topological Network Coverage	25 Junction Nodes · 39 Conduits (100% active)	All 25 nodes receive synchronized depth, risk classification, and serialization.

Regression Test Coverage

53 / 53 Tests Passing (100% Pass Rate)

Pytest suite verifying dry baseline, sag concentration, choke cascades, and routing.

Deterministic Offline Parity

Zero External Dependencies Required

Complete TypeScript client solver mirrors backend formulas for telecommunication outages.

 **PITCH SCRIPT & SPEAKER GUIDELINES (30 SECONDS):**

"Moving to Slide 3: Technical Approach. We built AGASTYA on a modern stack: React 19 and Leaflet on the frontend, FastAPI and NetworkX in Python on the backend. In our live validation tests, our physical surcharge model computes in under 5 milliseconds—allowing real-time slider manipulation from 0 to 100 mm/hr. Notice our verification metrics: 53 out of 53 unit and regression tests pass, including multi-node chokes and offline client parity. When the server goes down, the client-side engine continues to calculate routes without interruption."

SLIDE 4: FEASIBILITY, VIABILITY, SUPPORTING DATA & CHALLENGE MATRIX

Real-world feasibility facts, market size, civic viability, audience impact metrics, and challenge-solution engineering table.

- Zero Hardware Dependency:** Requires zero IoT water-level sensor installations to operate; functions instantly using open satellite DEM and street graphs.
- 100% Free & Open Datasets:** Leverages Open-Meteo precipitation API (no API key/billing), CartoDEM/SRTM elevation, and OpenStreetMap.
- High Scalability:** Deployable to any Indian municipal corporation in hours simply by supplying city bounding coordinates.
- Enormous Economic Addressability:** India suffers ₹14,000+ Crore in annual urban flood economic losses; over ₹25,244 Crore in public utility damages (2021).
- Emergency Dispatch Integration:** Plugs directly into existing Dial 108/112 ambulance CAD systems and municipal Smart City Integrated Command & Control Centers (ICCC).

Supporting Facts & Figures for Feasibility & Viability

Metric / Fact	Official Value	Official Source	Impact on AGASTYA Viability
Annual Flood Economic Loss	₹14,000+ Crore (\$14 Billion)	World Bank & NDMA Guidelines	Huge municipal ROI; preventing underpass inundation preserves commercial transit.
Public Utility Flood Damage	₹25,244 Crore in single year (2021)	Ministry of Home Affairs / Disaster Statistics	Provides ICCC operators proactive alerts before substations and roads flood.
Urban India Drainage Deficit	70%+ urban areas lack stormwater maps	CPHEEO / MoHUA Urban Reports	Proves market necessity of synthetic drainage generation

			(PySewer).
Monsoon Flood Fatalities	361 deaths (2024) · 2,754 (2019)	NCRB Accidental Deaths Report (ADSI)	Directly saves lives by stopping ambulances from deploying into submerged corridors.
Urban Addressable Market	4,000+ Urban Local Bodies (ULBs)	Smart Cities Mission / MoHUA	Scalable SaaS model at ₹5-10 Lakhs/yr/city yields ₹2,000+ Crore addressable market.
Emergency Travel Delay Saved	8 to 12 minutes saved per emergency	AIIMS / Delhi Traffic Police Studies	Enables ambulances to bypass blocked sag corridors during golden hour.

Challenge-Solution Engineering Matrix

Technical / Operational Challenge	Engineering Solution Implemented in AGASTYA	Verified Real-World Result
No public GIS drainage pipe blueprints in Indian cities	Integrated PySewer to synthesize gravity sewer networks from OSM centerlines and DEM slopes.	Generated realistic 25-node, 39-conduit network with zero manual CAD digitization.
Hydrodynamic 2D SWMM models take 30-60 mins to compute	Formulated 1D Manning pipe capacity coupled with rational overland surcharge on graph.	Sub-5 millisecond response time; supports real-time slider updates (<2 sec overall).
Telecom towers collapse during severe monsoon storms	Implemented dual-solver parity: full client-side TypeScript physical and Dijkstra engine.	100% functional offline demo; browser computes routes with zero backend connectivity.
Ambulances routed into flooded hospital access roads	Strict pre-Dijkstra endpoint validation with 15 cm water depth clearance threshold.	Returns structured ORIGIN_UNSAFE / DESTINATION_UNSAFE warnings; halts unsafe dispatch.
Debris and plastic clogs cause localized backwater flooding	Multi-node click-to-choke simulation engine with immediate hydraulic surcharge propagation.	Operators can simulate simultaneous choking of Minto Bridge and DDU Marg manholes.
Judges/operators confuse emergency destination with flood markers	Restructured map design tokens: Royal Indigo (#6366f1) destination vs Red (#ef4444) flood risk.	Visual collision eliminated; distinct target rings and tooltips prevent operational confusion.

 PITCH SCRIPT & SPEAKER GUIDELINES (30 SECONDS):

"On Slide 4, we prove feasibility and viability. Look at the numbers: India loses ₹14,000 Crore annually to urban floods, and 70% of Indian urban centers lack digitized stormwater blueprints. We don't ask municipalities to spend ₹50 Crore on IoT sensors. Our Challenge-Solution matrix highlights our biggest breakthroughs: we use PySewer to synthesize missing pipe networks, we replaced heavy 45-minute simulation runs with a 5-millisecond Manning solver, and our offline fallback ensures that when mobile networks fail, the local ambulance tablet continues to compute safe routes."

SLIDE 5: IMPACT, BENEFITS, UNIQUENESS & MARKET COMPARISON

Four-quadrant benefit analysis, comparison matrix vs market alternatives (✓/▲/×), and product-to-impact narrative story.

- **Social Benefits:** Preserves human life by ensuring critical ambulances never drown in underpass depressions; protects 10 Lakh+ daily commuters.
- **Economic Benefits:** Prevents vehicle submergence damage, saves millions in commercial transit disruption, costs <₹10L/city vs ₹2-5 Cr SCADA.
- **Technical Benefits:** Sub-second execution, hardware-agnostic, open-source compliance, dual online/offline solver parity.
- **Environmental Benefits:** Provides empirical drainage capacity data for sustainable stormwater upgrades, avoiding post-disaster reconstruction emissions.

Competitive Advantage vs Market Solutions

Key Capability	Traditional Flood Portals	Commercial Flood APIs	Municipal ICCC Dashboards	AGASTYA (Our Solution)
Street-Level Depth in Centimeters	× (City-wide only)	▲ (Inundation extent only)	× (Qualitative text)	✓ (Exact cm at all 25 nodes)
Underground Drainage Pipe Coupling	× (Ignored)	× (Overland only)	× (Static asset list)	✓ (Manning 1D pipe physics)
Synthetic Pipe Synthesis (No Data Needed)	× (Requires CAD)	× (Requires sensor feed)	× (Missing data problem)	✓ (Auto-generated via PySewer)
Real-Time Safe Ambulance Routing	× (No routing)	× (Historical maps)	▲ (Manual rerouting)	✓ (Automated Dijkstra <15cm)
Submerged Endpoint Safety Refusal	× (No safety checks)	× (No vehicle logic)	× (Unsafe dispatch)	✓ (ORIGIN / DESTINATION UNSAFE)
Multi-Node Choke Point Simulation	× (Static models)	× (No clog modeling)	× (Reactive complaints)	✓ (Interactive click-to-choke)
100% Offline Disaster Fallback	× (Crashes offline)	× (Requires cloud API)	× (Central server dependent)	✓ (Full client-side physical solver)
Total Cost of City Deployment	High (₹1–2 Cr consulting)	High (Recurring SaaS \$)	Extreme (₹5–20 Cr SCADA)	✓ (<₹10 Lakhs open software)

Product-to-Impact Case Study Narrative

 **Real-World Deployment Scenario: Monsoon Downpour at Minto Bridge Underpass**

- Time: 16:45 IST · Severe monsoon cloudburst (75 mm/hr) hits Central New Delhi.
- Baseline Danger: Minto Bridge sag elevation (210.5m) rapidly accumulates 38.3 cm of runoff within 20 minutes, becoming

- impassable.
- Emergency Call: An ambulance at Connaught Place Outer North receives an emergency cardiac dispatch to Lady Hardinge / Barakhamba.
 - Conventional Failure: Standard GPS routing directs the ambulance through the shortest geometric path directly under Minto Bridge, trapping the vehicle in waist-deep water.
 - AGASTYA Intervention: AGASTYA detects Minto Bridge sag at 38.3 cm (>15.0 cm limit), immediately prunes the flooded corridor, computes a dry arterial bypass in <2 milliseconds, and guides the driver via DDU Marg East—saving 11 minutes of golden-hour transit and preventing vehicle submergence.

 PITCH SCRIPT & SPEAKER GUIDELINES (30 SECONDS):

"Slide 5 presents our impact and market comparison. Compare AGASTYA with commercial flood APIs: competitors show satellite flood extents hours later. We predict exact centimeters in real time, couple underground pipe physics, and generate actionable routes. Listen to our product-to-impact story: during a 75 mm/hr downpour at Minto Bridge, standard navigation sends an ambulance straight into the 38 cm underpass. AGASTYA flags the underpass as impassable, calculates an alternate dry bypass in 2 milliseconds, and delivers the patient safely to the hospital."

SLIDE 6: RESEARCH, REAL REFERENCES & TEAM CONTRIBUTION

Empirical scientific citations, policy frameworks, dataset URLs, and individual team member task distribution.

Research Citations & Real Data References

Research Topic / Domain	Reference Citation & Official Resource	Key Empirical Data Point
Rainfall Climatology & Nowcasting	IMD Gridded Dataset (cdsp.imdpune.gov.in) · Open-Meteo API	High-resolution hourly precipitation data; validated against IMD Delhi rain gauges.
Topographical DEM Modeling	ISRO NRSC Bhuvan CartoDEM (bhuvan.nrsc.gov.in)	10-meter digital elevation model revealing 6m topographical gradient across Minto basin.
Synthetic Sewer Generation	PySewer: Automated Sewer Network Generation (JOSS 2024)	Peer-reviewed methodology for synthesizing gravity drainage networks in data-scarce regions.
Urban Drainage Engineering Code	CPHEEO Stormwater Drainage Manual (MoHUA)	Indian design standard conveyance: 18 mm/hr design intake, conduit slopes, Manning's n=0.015.
Open Street Network Modeling	OSMnx: Complex Street Networks (osmnx.readthedocs.io)	Topologically verified road network graphs extracted via OpenStreetMap Overpass API.
Economic Disaster Valuation	World Bank & Swiss Re 'Billion-Dollar Rain' Report (2024)	Quantifies \$14 Billion (₹14,000 Cr) annual flood loss baseline for urban India.
Urban Flood Disaster Policy	National Disaster Management Authority (NDMA) Guidelines	Direct alignment with NDMA mandate on urban flood nowcasting and hospital evacuation.

Mortality & Safety Benchmarks

NCRB Accidental Deaths & Suicides in India (ADSI 2024)

Empirical record of urban waterlogging casualties; validates 15 cm vehicle safety limit.

6-Member Team Contribution & Role Matrix

Member Name	Assigned Role	Core Responsibilities	Key Deliverables
Deepanshu (Lead)	Team Lead & System Architect	System design, full-stack integration, jury pitch delivery, technical defense.	Overall architecture, technical report, pitch script.
Member 2	Hydrologic & GIS Engineer	CartoDEM elevation sampling, PySewer pipe synthesis, Manning 1D calibration.	Topological graph, elevation gradients, pipe sizing.
Member 3	Frontend / UI/UX Lead	Leaflet map visualizer, operator dashboard, multi-node choke controls, glassmorphic design.	React 19 dashboard, color tokens, interactive popups.
Member 4	Algorithm & Routing Engineer	Dijkstra safe routing algorithm, vehicle clearance threshold gates, edge pruning.	Safe routing engine, endpoint safety validators, API endpoints.
Member 5	QA, Testing & Verification	Automated pytest suite, regression auditing, browser testing, offline verification.	53 passing unit tests, verification matrices, video recording.
Member 6	Research & Policy Analyst	Economic data collection, NDMA/CPHEEO compliance, municipal market analysis.	Facts sheet, competitive comparison table, slide deck content.

 PITCH SCRIPT & SPEAKER GUIDELINES (30 SECONDS):

"In conclusion on Slide 6: our entire solution is grounded in authentic, peer-reviewed research and Indian governmental standards—from CPHEEO drainage codes to ISRO CartoDEM and IMD radar grids. Our 6-member team brings together hydrology, graph algorithms, and geospatial UI engineering. AGASTYA is not a concept or a Figma mockup; it is a fully functional, verified prototype ready to deploy for New Delhi and beyond. Thank you!"